NAZAR v5 - Circuit Diagrams

ESP32 DevKit (38-pin) - 24 V adapter, two LM2596 bucks, belt relay for stop-and-inspect - 1 Nov 2026

Power

From	To	Notes
24 V adapter +	LM2596 #1 IN+ and LM2596 #2 IN+	adapter 2 A or more
24 V adapter -	LM2596 #1 IN- and LM2596 #2 IN-	common ground
LM2596 #1 OUT+	ESP32 5V pin, MG996R red, C1 +, relay DC+	set to 5.5 V
LM2596 #1 OUT-	ESP32 GND, MG996R brown, C1 stripe, relay DC-	
LM2596 #2 OUT+	LED strip +12V, S1 -> INA219 VIN+	set to 12 V
INA219 VIN-	Relay NO	INA219 measures motor current
Relay COM	Belt motor +	belt runs only while the relay is on
LM2596 #2 OUT-	LED strip GND, belt motor -	

ESP32 pin map

ESP32 pin	Goes to	Notes
GPIO21 (SDA)	INA219, MPU-6050, BH1750 SDA	blue
GPIO22 (SCL)	INA219, MPU-6050, BH1750 SCL	green
3.3 V	I2C modules, FC-51, DS18B20, 4.7 kΩ	red
GPIO18	FC-51 OUT	purple
GPIO4	DS18B20 DQ + 4.7 kΩ to 3.3 V	yellow
GPIO13	330 Ω -> MG996R signal	cyan
GPIO26	Relay IN (stops the belt under the camera)	indigo
5V	LM2596 #1 OUT+ (5.5 V)	orange
GND	all grounds joined	black

Before you power it

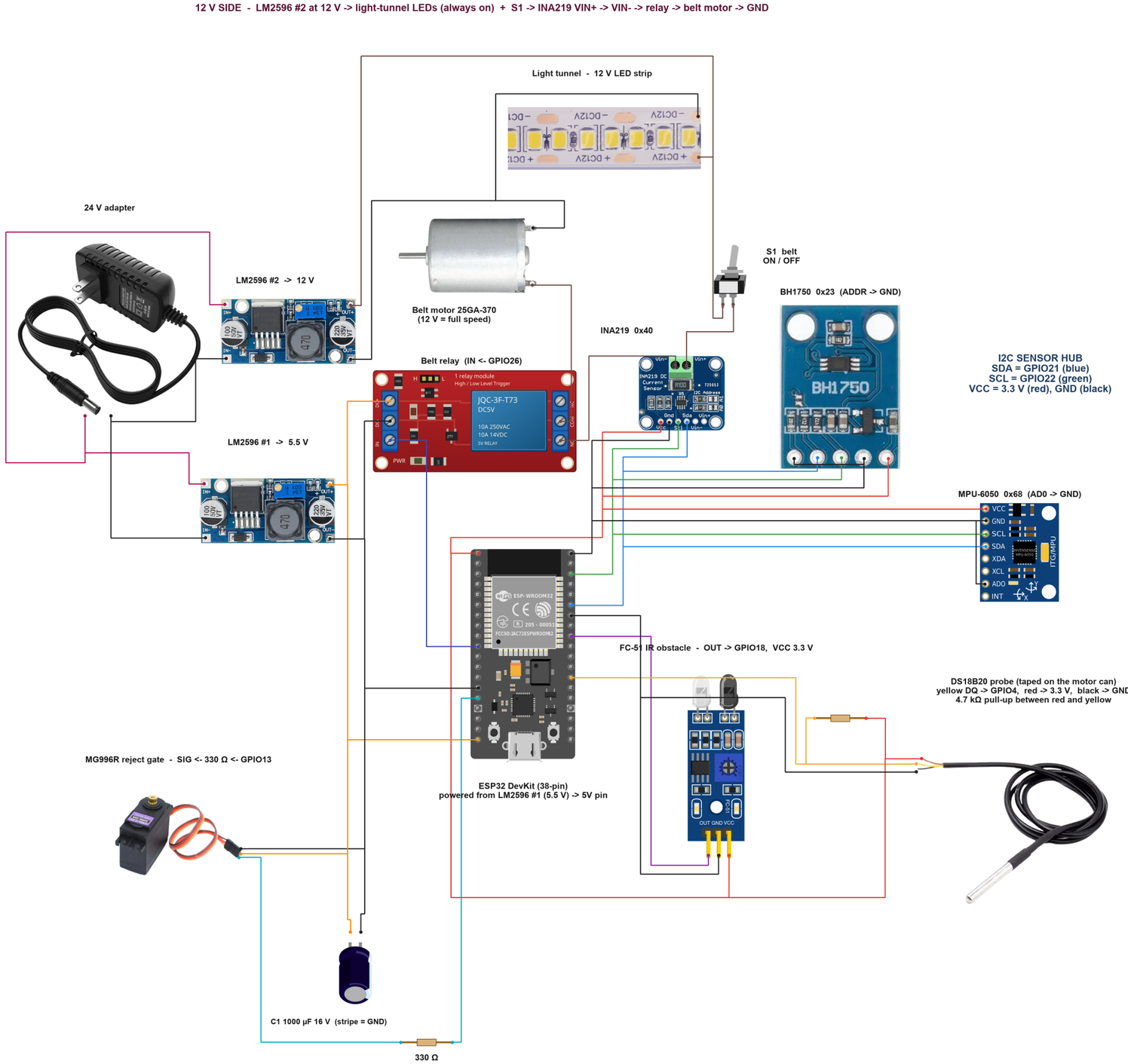
1. Connect only the 24 V adapter. Turn each buck's brass screw and read the output with a multimeter: LM2596 #1 = 5.5 V, LM2596 #2 = 12 V.
2. Mark the bucks "5V" and "12V" with tape. 12 V on the ESP32 side destroys the ESP32 and the servo.
3. Fit C1 (1000 µF 16 V) across the 5.5 V rail: + to 5.5 V, stripe to GND.
4. Use thick wire (about 20 AWG) for both power rails, and stick a small heatsink on LM2596 #1 - the servo kick pulls high current.
5. If the ESP32 resets when the servo moves, add another 470-1000 µF capacitor right at the ESP32 5V and GND pins.
6. Check the ESP32 has its USB diode before plugging USB in while the 5.5 V rail is live, or unplug the 5V wire while uploading code.
7. Relay: set its H/L jumper to L (low trigger, the firmware default). The belt only runs while the ESP32 is on and holds the relay; it stops under the camera for each washer. A 1N4007 across the motor (stripe to +) is a plus.

I2C addresses

I2C device	Address	Strap
INA219	0x40	default
MPU-6050	0x68	AD0 -> GND
BH1750	0x23	ADDR -> GND

Complete wiring

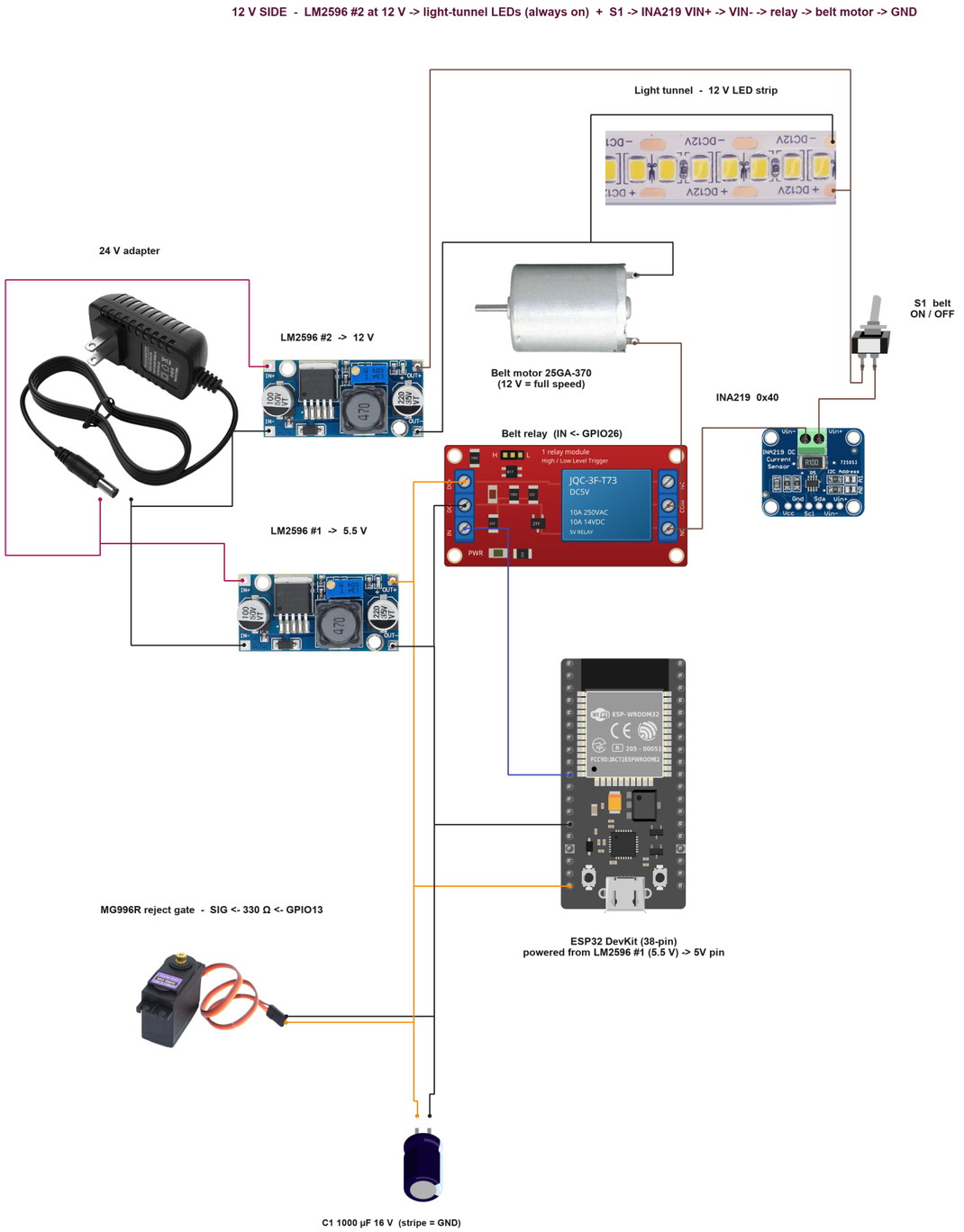
Everything on one sheet. 24 V adapter -> LM2596 #1 (5.5 V) and LM2596 #2 (12 V).



POWER - 24 V adapter feeds two LM2596 bucks. #1 at 5.5 V (orange) -> ESP32 5V pin + MG996R. #2 at 12 V (brown) -> light-tunnel LEDs + S1 / INA219 / relay / belt motor. Relay coil on #1, IN = GPIO26 (indigo). Both bucks turned 180° (IN left). All GND joined (black).

Part 1 - Power, light tunnel, belt motor and relay

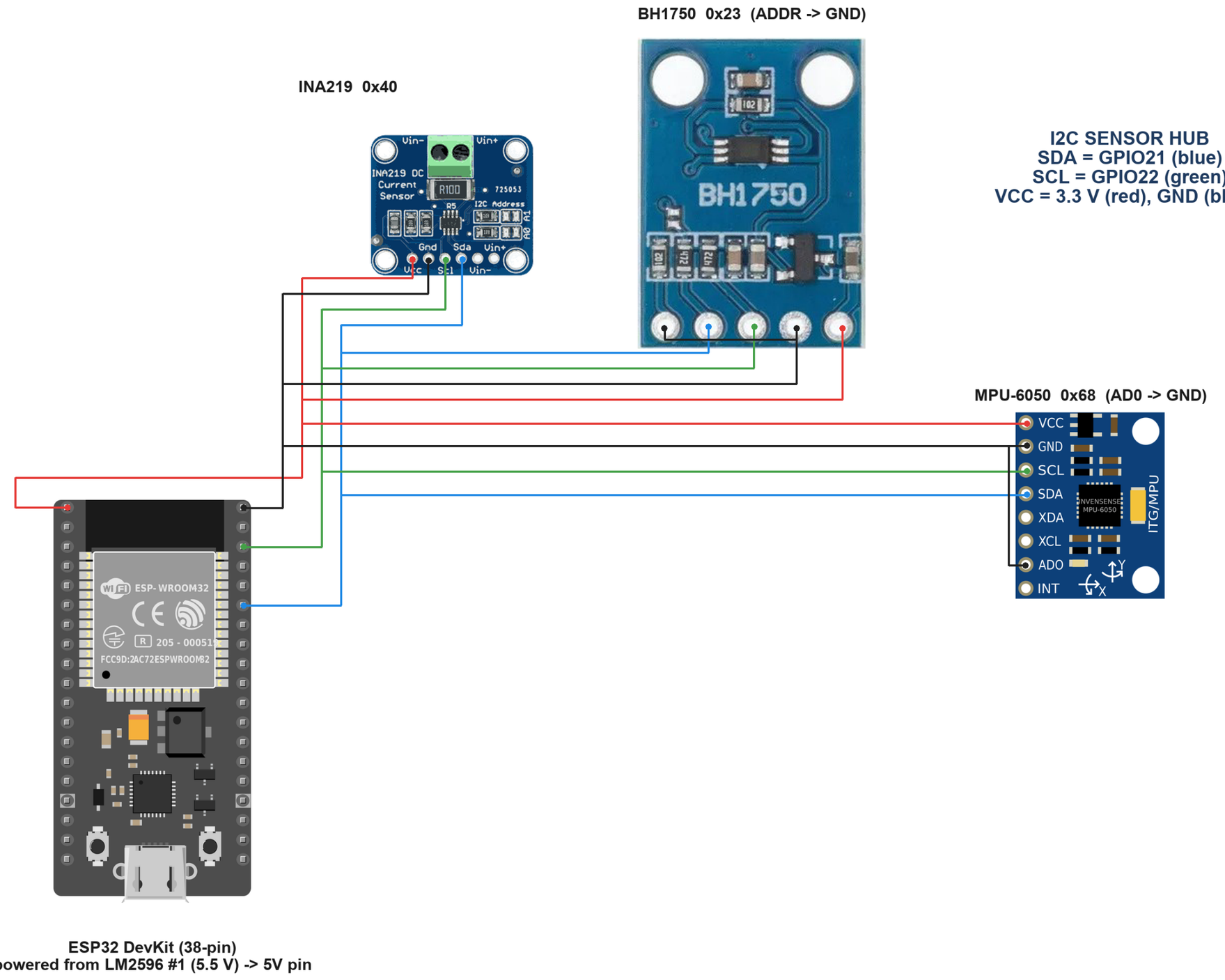
LM2596 #1 (5.5 V): ESP32 5V pin, MG996R, C1, relay coil. LM2596 #2 (12 V): LED strip (always on) and S1 -> INA219 -> relay -> belt motor. Relay IN = GPIO26.



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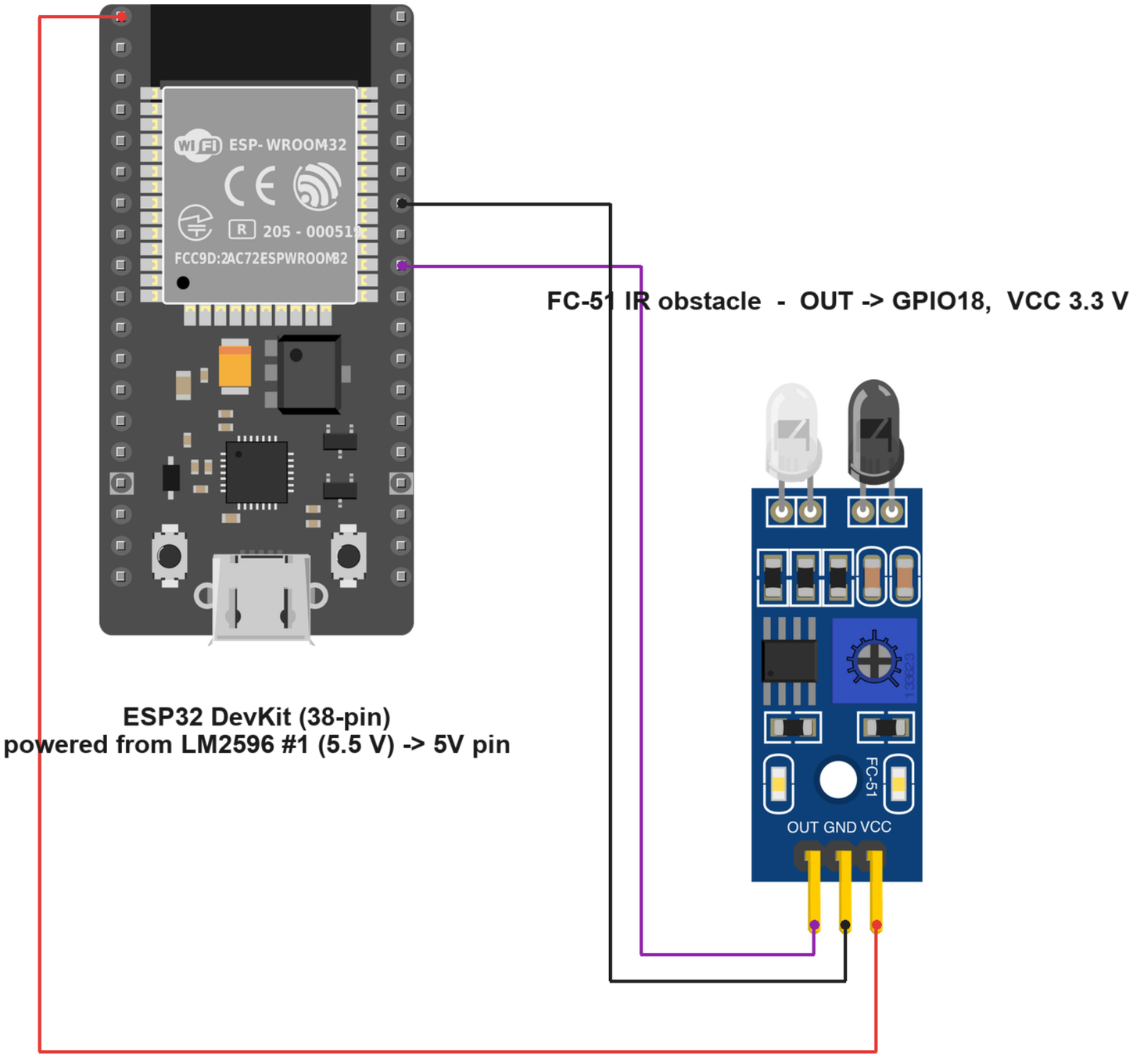
Part 2 - I2C sensor hub

INA219 (0x40), MPU-6050 (0x68, AD0 to GND), BH1750 (0x23, ADDR to GND) on SDA = GPIO21, SCL = GPIO22, 3.3 V.



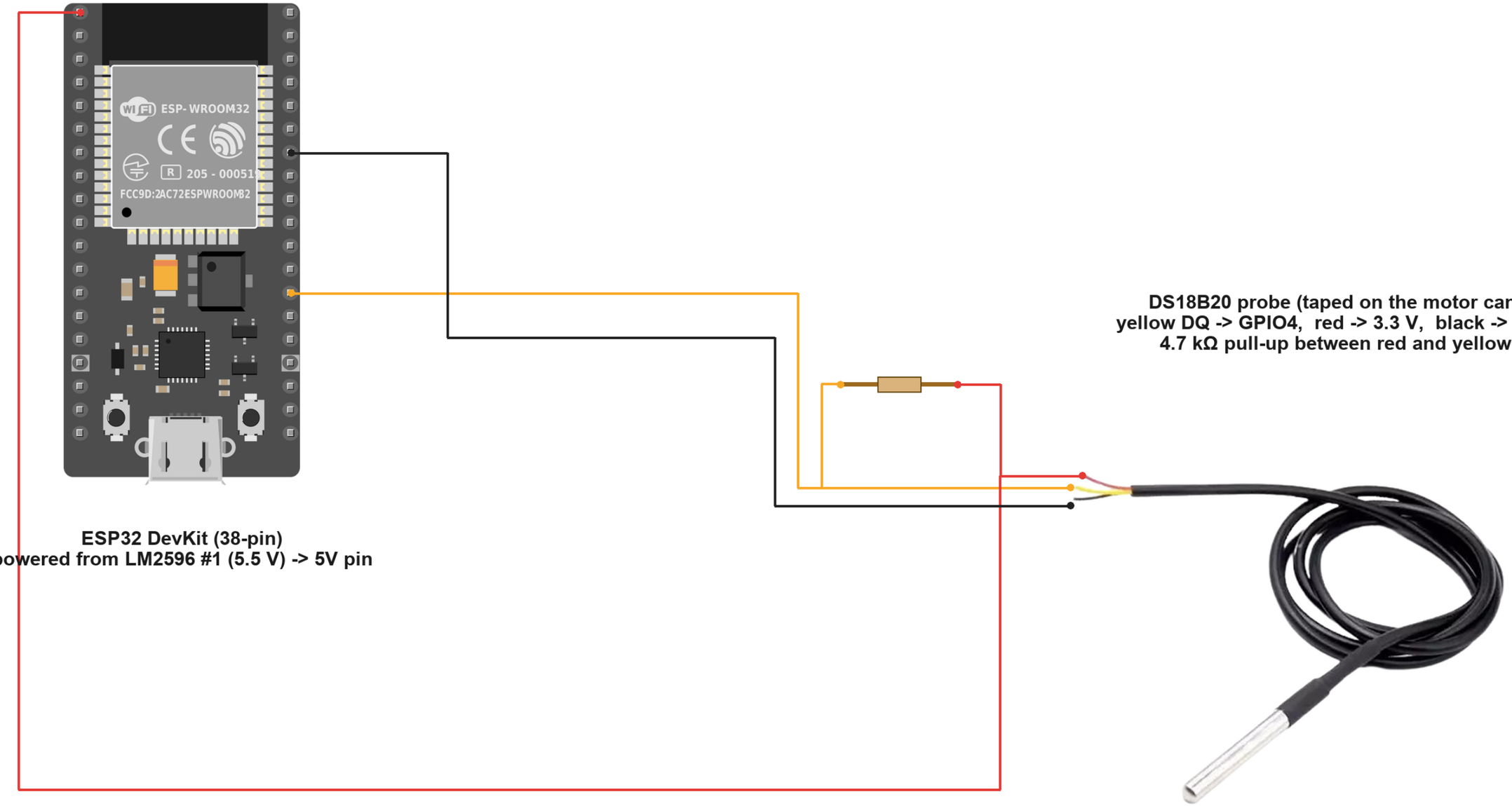
Part 3 - FC-51 IR obstacle sensor

OUT -> GPIO18. Runs on 3.3 V so the output is safe for the ESP32.



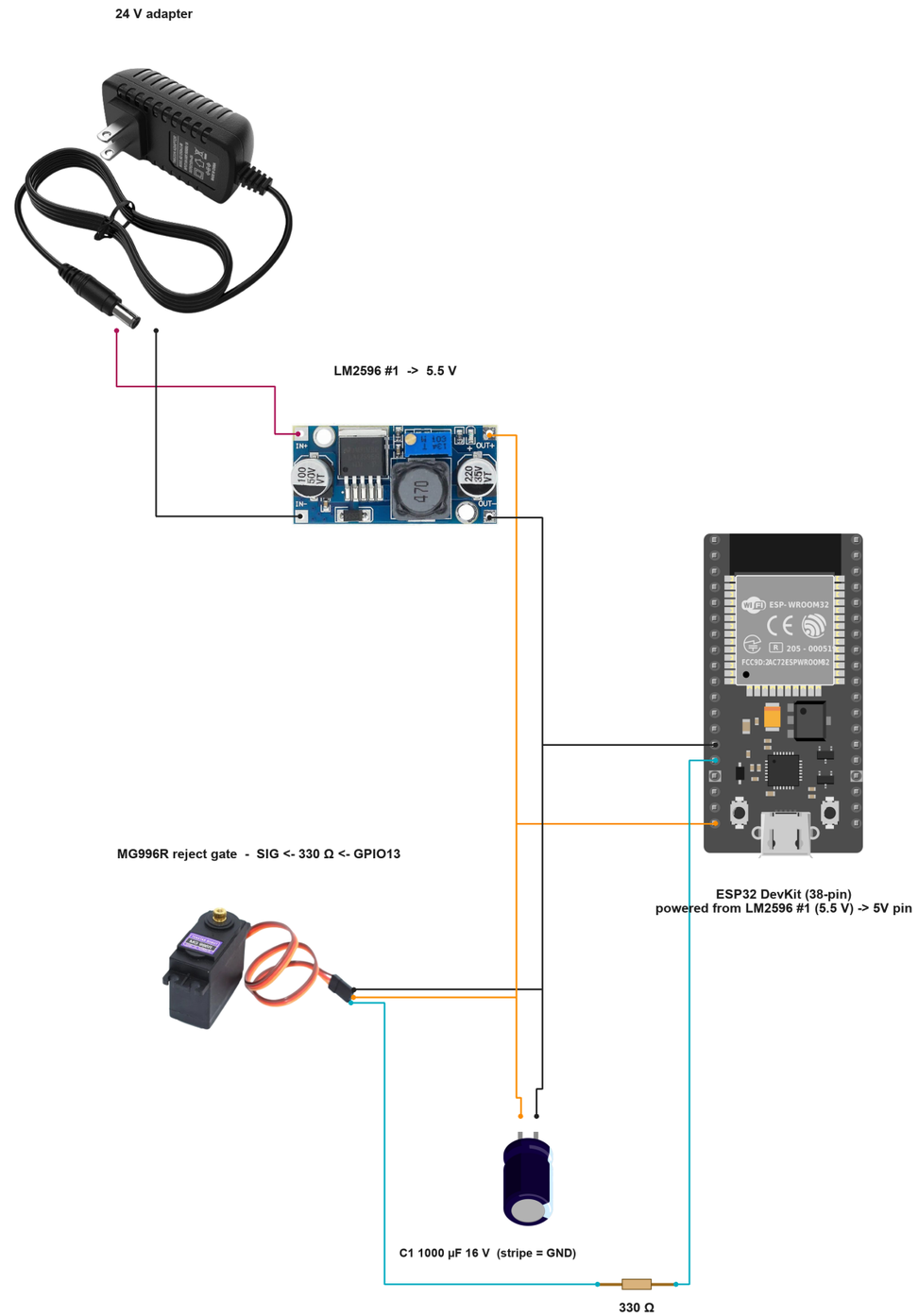
Part 4 - DS18B20 motor temperature

Yellow DQ -> GPIO4, red -> 3.3 V, black -> GND. 4.7 kΩ pull-up between red and yellow.



Part 5 - Reject gate servo

MG996R powered from LM2596 #1 (5.5 V). Signal from GPIO13 through a 330 Ω resistor.



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